

Instructions:

The SOC at BulbaTech Innovations received an alert of abnormal traffic patterns and a high number of repeated queries originating from one of their external-facing endpoints (172.16.1.16).

Using your newfound Wireshark skills, perform network traffic analysis on the wireshark_challenge.pcap packet capture to produce findings for the report questions below.

Challenge File:

02_Network_Security/02_Wireshark/Challenges/wireshark_challenge.pcap

Q1 How many total packets are in the wireshark_challenge.pcap packet capture?

39106

Q2 What was the first domain name queried and resolved in the capture?

webmasterdev.com

Q3 What is the associated IP address of the domain name?

184.168.98.68

Q4 How many HTTP packets are contained in the capture file?

8

Q5 What is the relative path the victim accessed on the web server to request a file for download?

/9GQ5A8/6ctf5JL

Q6 Based on the response header, what file type or format does the web server claim the downloaded file to be?

gif

Q7 However, what is the actual file signature or magic bytes contained in the file?

MZ

Q8 What command-line utility or program was used by the victim to download the file?

WindowsPowerShell

Q9 What is the sha256 hash of the downloaded file?

9b8ffdc8ba2b2caa485cca56a82b2dcdbd251f65fb30bc88f0ac3da6704e4d3c6

Q10 Submit the uncovered hash to VirusTotal. Based on the popular threat label and tags, what type of malware did the endpoint get infected with?

pikabot

Q11 What protocol makes up the majority of UDP packets?

dns

Q12 Look at the domain names that were queried within the capture. In defanged format, what is the base domain name that is continually queried?

dns[.]steasteel[.]net

Q13 Read up on MITRE ATT&CK ID T1071.004. What is the attack technique we're likely seeing in the PCAP file often known as?

DNS Tunneling

Challenge URL

<https://challenges.malwarecube.com/#/c/7265ec1c-9773-4c7c-9ed4-2ea26e19f346>

PCAP Provided By

<https://www.malware-traffic-analysis.net/>